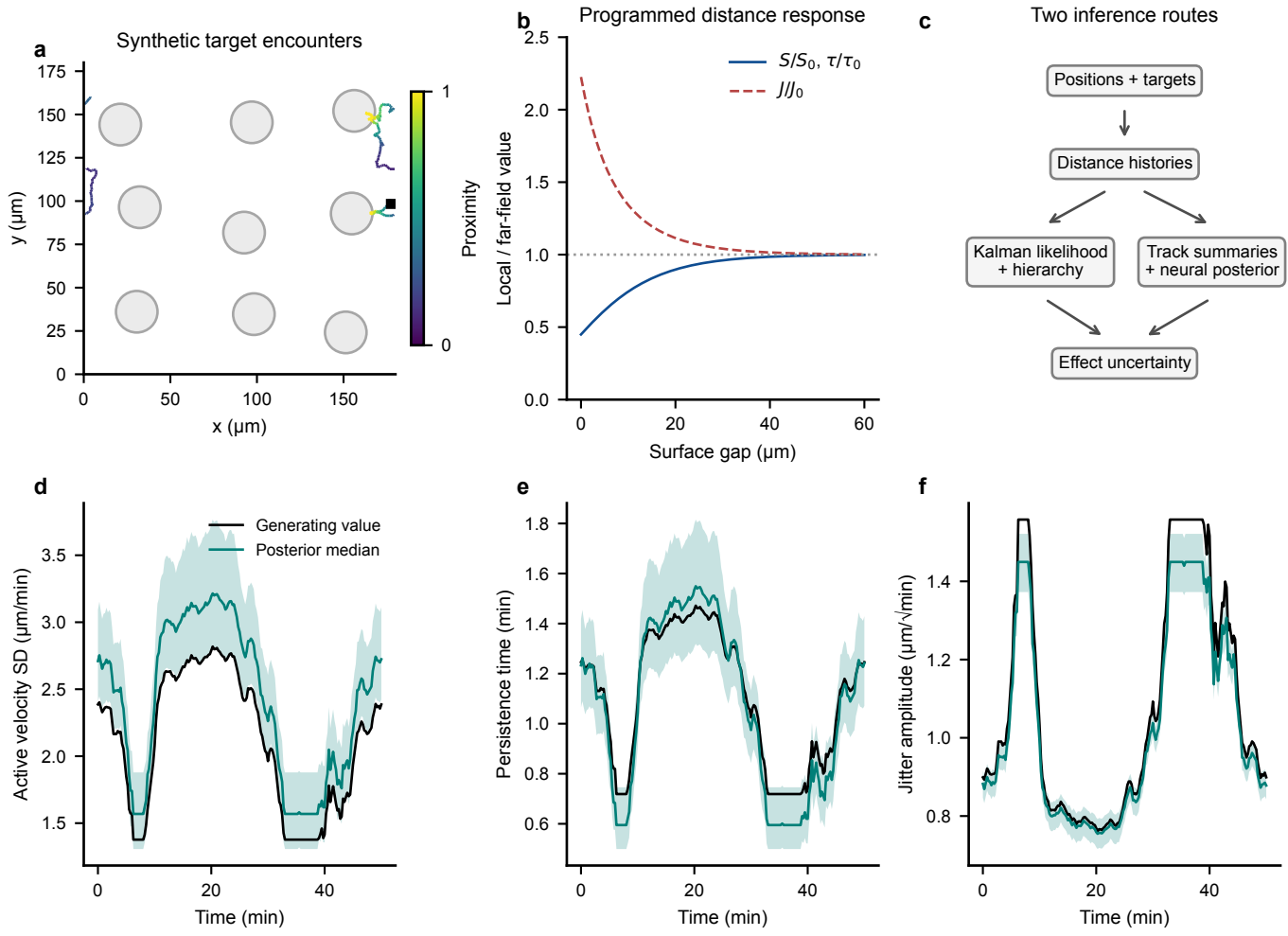

a
Synthetic target encounters
y (µm)
x (µm)
Proximity
1
0
b
Programmed distance response
S/S_0, τ/τ_0
J/J_0
Local / far-field value
Surface gap (µm)
c
Two inference routes
Positions + targets
Distance histories
Kalman likelihood + hierarchy
Track summaries + neural posterior
Effect uncertainty
d
Generating value
Posterior median
Active velocity SD (µm/min)
Time (min)
e
Persistence time (min)
Time (min)
f
Jitter amplitude (µm/√min)
Time (min)